

Algorithmic Containment: Directionality, Uniformity, and a Self-Delimiting Repair

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Abstract

Franz’s proposal to understand implication through containment between generative descriptions gives a promising non-probabilistic world-model semantics for inference. We report a source-level and machine-checked audit of the two principal results in the 2026 formulation. The displayed proof of Theorem 1 reverses a directional mutual-information premise and later uses a feature-compression hypothesis absent from its theorem statement. The directional step can be repaired with an explicit symmetry-of-information loss, while the compression premise should be stated. The construction in Theorem 2 asks a program to recover a feature from $\langle g^*, K(g^*) \rangle$, but its encoded parameter transports the program and residual without transporting the exact value $K(g^*)$. We give a uniform self-delimiting repair carrying the triple $(K(g^*), t, p)$ and prove its extraction, reconstruction, injectivity, and finite slack ledger in Lean.

These findings invalidate the two displayed proof routes, not the research programme. The corrected construction establishes exact image inclusion and additive-slack feature transfer. Completing the fully universal prefix-complexity theorem still requires an effective conditional prefix machine, the strong conditional chain rule, and symmetry of information with uniform constants. We state that boundary explicitly and invite correction of any source interpretation in this preliminary note.

1 Scope, conventions, and attribution

The theory audited here is due to Arthur Franz, especially *Grounded Reasoning: Implication as Algorithmic Containment* [1], building on the incremental-compression theory of Franz, Antonenko, and Soletskyi [2] and the earlier programme [3]. The prefix-complexity chain rule and symmetry results are standard algorithmic-information theory; our reference point is Li and Vitányi [4].

The following distinctions are essential.

- $K(a \mid b)$ is prefix Kolmogorov complexity relative to one fixed universal prefix machine.
- The 2026 paper defines directional information by $I(a : b) = K(a) - K(a \mid b)$. It is not definitionally symmetric.
- An $O(1)$ or $O(\log n)$ statement is meaningful here only as a uniform family statement: its hidden constant cannot depend on the particular strings being bounded.
- Exact featurehood, requiring $|f| + |p| < |x|$, is different from featurehood with an additive logarithmic slack. Our formalization gives them different types.

This note distinguishes three judgments: a displayed proof may be invalid; a theorem statement may require an omitted premise; and an existential theorem may remain true by a repaired construction. We do not infer the last judgment from either of the first two.

2 Theorem 1: the directional step

Theorem 1 of [1] assumes, in the paper’s notation,

$$f(p) = x, \tag{1}$$

$$K(g \mid x) = O(\log|x|), \tag{2}$$

$$I(p : f, g) = O(1), \tag{3}$$

and concludes $K(g \mid f) = O(\log|x|)$.

With the paper’s directional definition, its third premise expands as

$$K(p) - K(p \mid f, g) = O(1).$$

Equation (16) of the proof instead uses

$$K(f, g \mid p) = K(f, g) + O(1).$$

That is the reverse orientation. It does not follow by unfolding the stated definition. A finite conditional prefix-machine example in the formal audit realizes the two directional values as exactly zero and one bit, so the swap is not a definitional equality.

The appropriate repair is not to declare mutual information exactly symmetric. Rather, one introduces the explicit symmetry ledger

$$K(f, g) + K(p \mid f, g) \leq K(p) + K(f, g \mid p) + s_{\text{sym}}, \tag{4}$$

where s_{sym} is logarithmic under the usual size controls. Combining the source-oriented genericity inequality

$$K(p) \leq K(p \mid f, g) + s_{\text{gen}}$$

with (4) gives the orientation actually used later:

$$K(f, g) \leq K(f, g \mid p) + s_{\text{gen}} + s_{\text{sym}}.$$

This cancellation is a theorem over natural numbers; no signed subtraction is needed in the proof ledger.

2.1 The omitted compression premise

The displayed assumptions of Theorem 1 do not include

$$|f| + |p| < |x|. \tag{5}$$

The proof nevertheless invokes (5) after its equation (19) to infer $|f| < |x|$ and hence to bound the cost of transporting $K(f)$. The premise belongs in the theorem statement. It is available in Corollary 1 because that corollary assumes that f is a feature of x , but it is absent from the more general theorem as displayed.

2.2 Uniform corrected form

The machine-checked repair proves the following finite statement. Assume:

1. explicit programs for $f(p) = x$ and for recovering g from x ;
2. representability of every complexity term used in a shortest-program argument;
3. fixed, condition-uniform simulations of application, projection, literal, indexing, and pairing machines;
4. a strong conditional chain rule with one uniform additive constant;
5. the direct genericity inequality, or the source inequality together with an explicit symmetry ledger; and
6. the compression premise (5).

Then

$$K(g \mid f) \leq K(g \mid x) + 2\log_2(|x| + 1) + C,$$

where C is displayed as a sum of the fixed compiler, chain-rule, genericity, and symmetry slacks. The formal proof uses only natural-number inequalities. The strong chain rule and symmetry theorem are currently interfaces rather than constructed instances; consequently this is a proved reduction of Theorem 1 to standard effective AIT infrastructure, not yet a closed construction of that infrastructure.

A subtle uniformity point matters here. A machine that hard-wires particular conditions can establish an inequality for those particular strings, but its compiler length may grow with those strings. Such a machine cannot witness a family-uniform $O(1)$. The corrected formal proof therefore uses fixed projection machines whose code is independent of f, p, g , and x .

3 Theorem 2: the missing complexity index

Theorem 2 chooses a shortest program t witnessing

$$|t| = K(f \mid g^*, K(g^*)).$$

Thus t expects both g^* and the exact integer $K(g^*)$ as auxiliary input. The construction then sets

$$q = E_2(t)p$$

and says that a fixed interpreter reconstructs f from (g^*, t) . The encoded parameter contains t and p , but not $K(g^*)$. In general the missing integer is not computable from g^* ; supplying it by a program whose length is uniform in g^* is precisely what cannot be assumed.

This is a gap in the displayed construction. It does not by itself disprove the existential conclusion of Theorem 2.

3.1 A self-delimiting repair

Let $k = K(g^*)$, and encode

$$q' = E_2(E_2(\text{bin}(k))t)p.$$

Equivalently, q' is a nested self-delimiting encoding of the triple (k, t, p) . A single fixed wrapper now:

1. decodes k, t, p from q' ;
2. runs t with condition $\langle g^*, \text{bin}(k) \rangle$ to obtain f ; and
3. runs f on p .

No complexity oracle is used. The wrapper is uniform in all four strings. The corresponding descriptive program first uses the old descriptive program to recover p from the source and then emits the same triple (k, t, p) . Hence extraction and reconstruction are both executable, rather than merely an image-membership assertion.

The exact residual length is

$$|q'| = |E_2(\text{bin}(k))t| + 2|\text{bin}(|E_2(\text{bin}(k))t|)| + 1 + |p|. \quad (6)$$

The formalization proves this identity and the logarithmic upper bound. It also proves the necessary negative control: the map that transports only (t, p) is not injective as an encoding of triples (k, t, p) , since it erases the first coordinate.

3.2 What is proved, and what remains

Let $G(g^*)$ be the tagged wrapper generator and let $H(k, t) = E_2(\text{bin}(k))t$. Suppose the finite description-transfer inequality

$$|G(g^*)| + |H(k, t)| \leq |f| + s_{\text{transfer}} \quad (7)$$

holds. The executable Lean theorem proves that every strict f -feature witness $f(p) = y$ becomes a $G(g^*)$ -feature witness with slack

$$s_{\text{transfer}} + 2|\text{bin}(|H(k, t)|)| + 1.$$

The same fixed generator works for every such p . Therefore the repaired construction proves exact image inclusion and a universal additive-slack feature implication. A concrete finite positive instance is included.

Two boundaries should not be blurred.

- The result is additive-slack featurehood, not the strict featurehood defined by $|g| + |q| < |y|$. The earlier theory already notes a logarithmic slack, and the formal types preserve that distinction.
- Deriving (7) from the source's prefix-complexity premises still depends on the strong chain rule and the shortest- feature bounds. The 2021 incompressibility theorem supplies the one-sided estimate $|g^*| \leq K(g^*) + O(1)$ needed in that ledger. The equality written as equation (29) in the 2026 proof is stronger than the cited theorem, but the stronger direction is unnecessary.

4 Machine-checked artifact map

The audit is organized by concept rather than by author or paper:

Module	Role
KolmogorovComplexity. SelfDelimitingCode	E_1/E_2 round trips, prefix-free ranges, exact and logarithmic lengths, and a negative comparison with unary headers.
KolmogorovComplexity. DirectionalInformation	Finite machine separating the two directional information values.
KolmogorovComplexity. ContainmentRepair	Fixed machine transformations, explicit Nat ledgers, and corrected direct- and source-orientation forms of Theorem 1.
WorldModel.ContainmentEncoding	Indexed triple residual, decoder round trip, injectivity, length bound, and the index-erasure counterexample.
WorldModel. FeatureContainmentConstruction	Executable wrapper, cross-algorithm implication, exact slack transfer, and a nonvacuous finite positive instance.

All listed crown theorems are free of admitted proofs and new axioms. Their reported dependencies are the ordinary logical foundations already used by Lean and Mathlib. The interfaces for the effective strong chain rule, universal prefix simulation, and full symmetry theorem remain visibly undischarged; no theorem in the artifact pretends otherwise.

5 Assessment

The source issues are substantive but localized.

Claim	Audit status
Theorem 1 proof as printed	Invalid at equation (16); it also uses an unstated compression premise.
Corrected Theorem 1	Reduced in Lean to fixed machine simulations, the strong conditional chain rule, and an explicit symmetry bound.
Theorem 2 construction as printed	Incomplete because the condition $K(g^*)$ needed by t is not transported.
Repaired image inclusion	Proved operationally using the indexed residual $(K(g^*), t, p)$.
Repaired compression transfer	Proved with an exact finite slack from an explicit description-transfer inequality; the remaining AIT derivation of that inequality is open.

The strongest surviving idea is also the most useful for world-model semantics: a generative premise and residual jointly produce a world, and a consequence is model-level only when it is stable under admissible residual variation rather than smuggled in by one residual. The repaired formalization makes that idea proof-relevant. It therefore supports continued development of algorithmic containment while separating recoverability, uniformity, strict compression, logarithmic slack, and efficient discoverability.

References

- [1] A. Franz, “Grounded Reasoning: Implication as Algorithmic Containment,” in *Artificial General Intelligence (AGI 2026)*, LNAI 16854, pp. 238–252, 2026. doi:10.1007/978-3-032-33010-9_15.
- [2] A. Franz, O. Antonenko, and R. Soletskyi, “A Theory of Incremental Compression,” *Information Sciences*, vol. 547, pp. 28–48, 2021. doi:10.1016/j.ins.2020.08.035.
- [3] A. Franz, “Some Theorems on Incremental Compression,” in *Artificial General Intelligence (AGI 2016)*, pp. 74–83, 2016.
- [4] M. Li and P. Vitányi, *An Introduction to Kolmogorov Complexity and Its Applications*, 4th ed., Springer, 2019.